HACKATHON EXETER

26 MAY 2026

CHRISTOPHER CARR

OVERVIEW

00: BLOCKCHAIN GAME

01: IMPLEMENTATION

02: QUANTUM

SCHEDULE

- 09:45 INTRODUCTION
- 10:00 THE BLOCKCHAIN GAME
- BREAK
- 11:20 REFLECTIONS ON THE HACKATHON
- 11:30 BUILDING A BLOCKCHAIN, DESIGN AND IMPLEMENTATION
- 12:15 QUANTUM AND POST QUANTUM IN BLOCKCHAINS AND MORE
- LUNCH

TOOLS

- Pen
- Paper
- VSCode
- Github
- Unix/Linux Bash
- AI

https://github.com/ca-carr/hackathon-EXE-25-26

IDEA

Main idea is to build our own understanding:

- With research;
- And ai;
- to understand;
- and to build blockchain infrastructure.

We will be using this Github page below as a starter https://github.com/ca-carr/hackathon-EXE-25-26

LEARNING GOALS

1. Understand Blockchain from a deeper, conceptual level
2. Develop your knowledge and familiarity with AI tools for development, coding and understanding
3. Gain experience using tools, and developing our own understanding and knowledge
4. Gain some insight into quantum threats and post quantum ideas

GAME #1

THE LOCAL BLOCKCHAIN

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GROUPS & TEAMS

A

SENDERS & RECEIVERS

2 SENDER TEAMS
of 4 groups

2 groups per team

B

NETWORK & ADVERSARIES

1 NETWORK TEAM
of 2 groups

2 groups to one team

C

VERIFIERS & RECORDERS

5 VERIFIER TEAMS
of 5 groups

Each group their own team

USE YOUR BRAINS, PEN AND PAPER: NO PHONE / NO LAPTOP

GAME IS TO SIMULATE A BLOCKCHAIN WITH PAPER & PEN

ROUND 1:

- 15 minutes setup time (decide on the protocol)
- 20 minute game (play the blockchain game)

ROUND 2:

- 5 minutes improve rules (change parties)
- 20 minute game

PART 01
BUILDING BLOCKCHAINS

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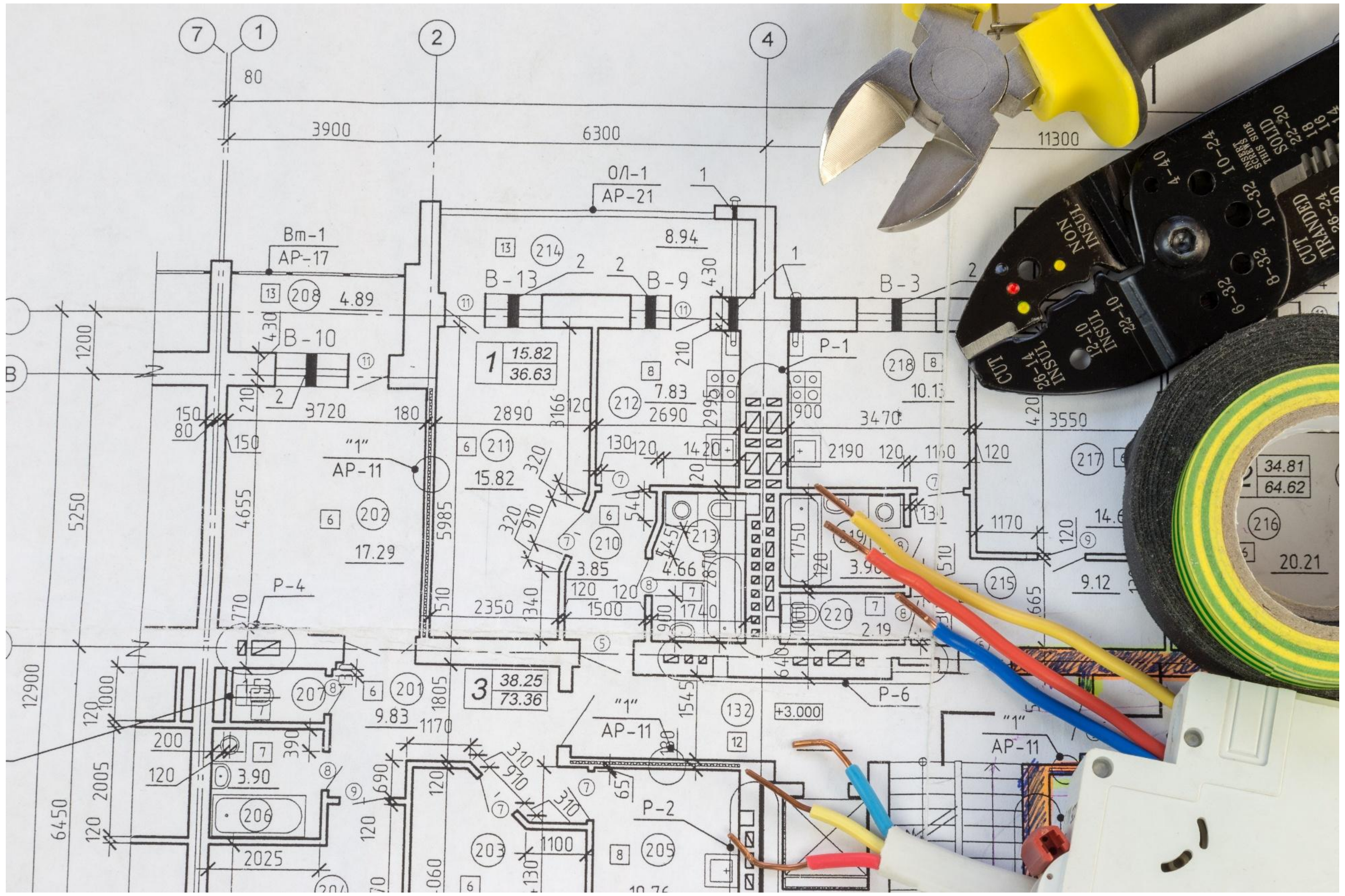
BUILDING BLOCKCHAINS

Say we want to construct our own blockchain

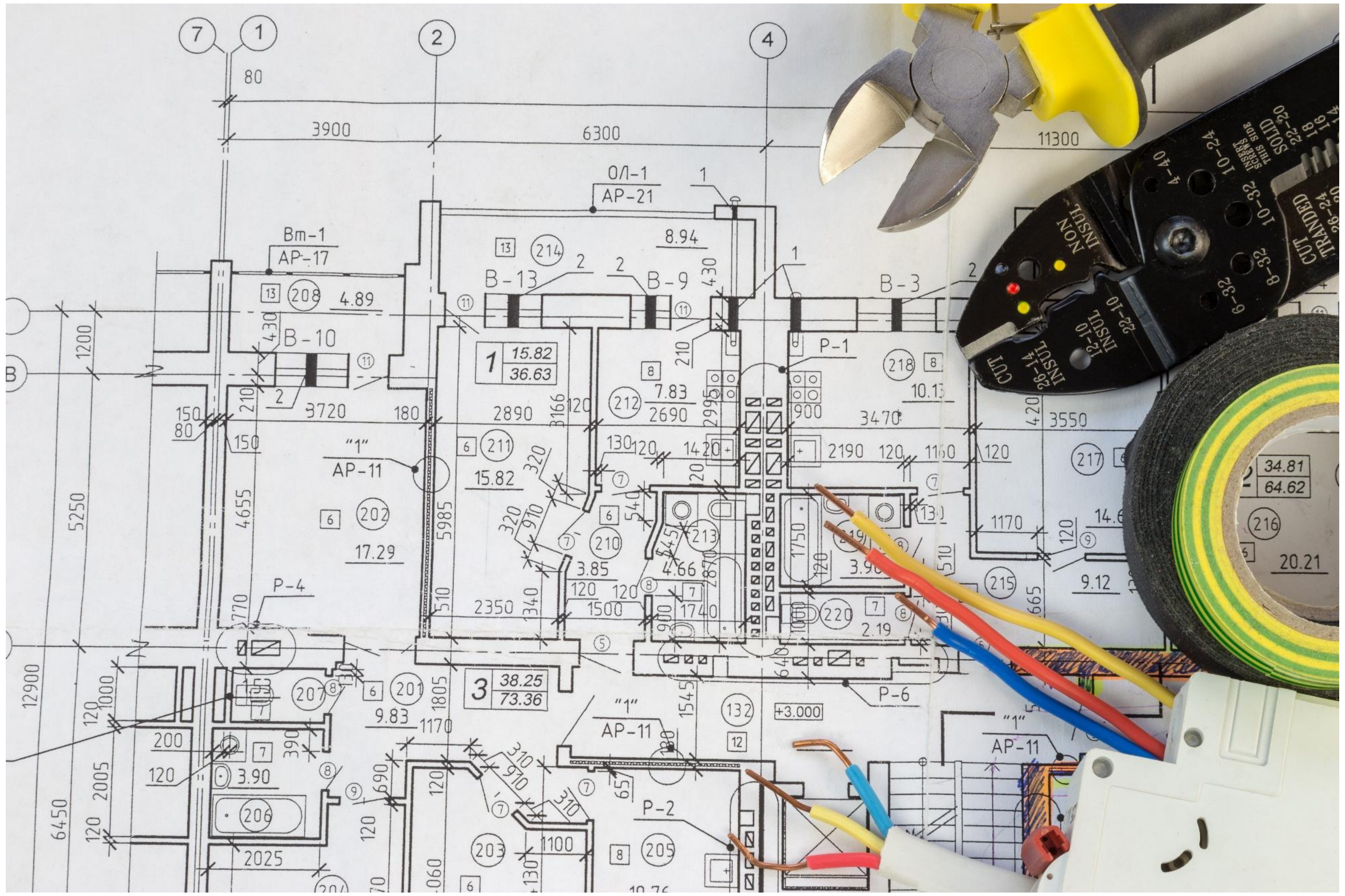
What aspects to we need to consider?>>>

ASPECTS



- The Network
- Verification
- Updating State
- Permission
- Agreement
- Global Agreement

ASPECTS



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THE NETWORK

- Q: WHY ARE BLOCKCHAIN NETWORKS DIFFICULT?

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- Timing, P2P, Peer-discovery, Flooding, DoS / DDoS, Inactivity, Resource usage, utilization

THE NETWORK

Building the network infrastructure is one of the most challenging aspects of blockchain design.

We are going to deploy our own (simple) blockchain network, for which we can communicate with peers, essentially building the Network aspect from Part 1.

THE NETWORK

We will be following the examples on the GitHub.

In small groups, we will set up a p2p-network that forwards messages around the network and connects to other clients on the network.

Once we have that working, we will then add on

- transactions,
- verification
- and storage.

THE NETWORK: MILESTONES & WINNING

1. The first group to get their blockchain network setup.
2. The first group to send and receive a transaction on their network.
3. First two groups to get one-another to connect to their network(s).

PART 02

Q U A N T U M

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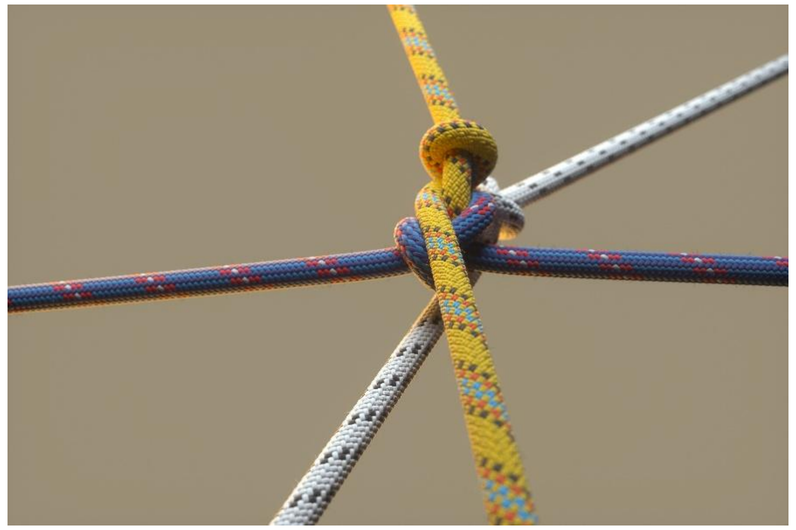
QUANTUM

- Quantum machines use Qbits instead of bits.
- They use *physical* properties, such as entanglement, to achieve exponential speed-ups for solving specific problems
  - such as integer factorization (RSA)
  - and discrete logarithms (ECDSA)

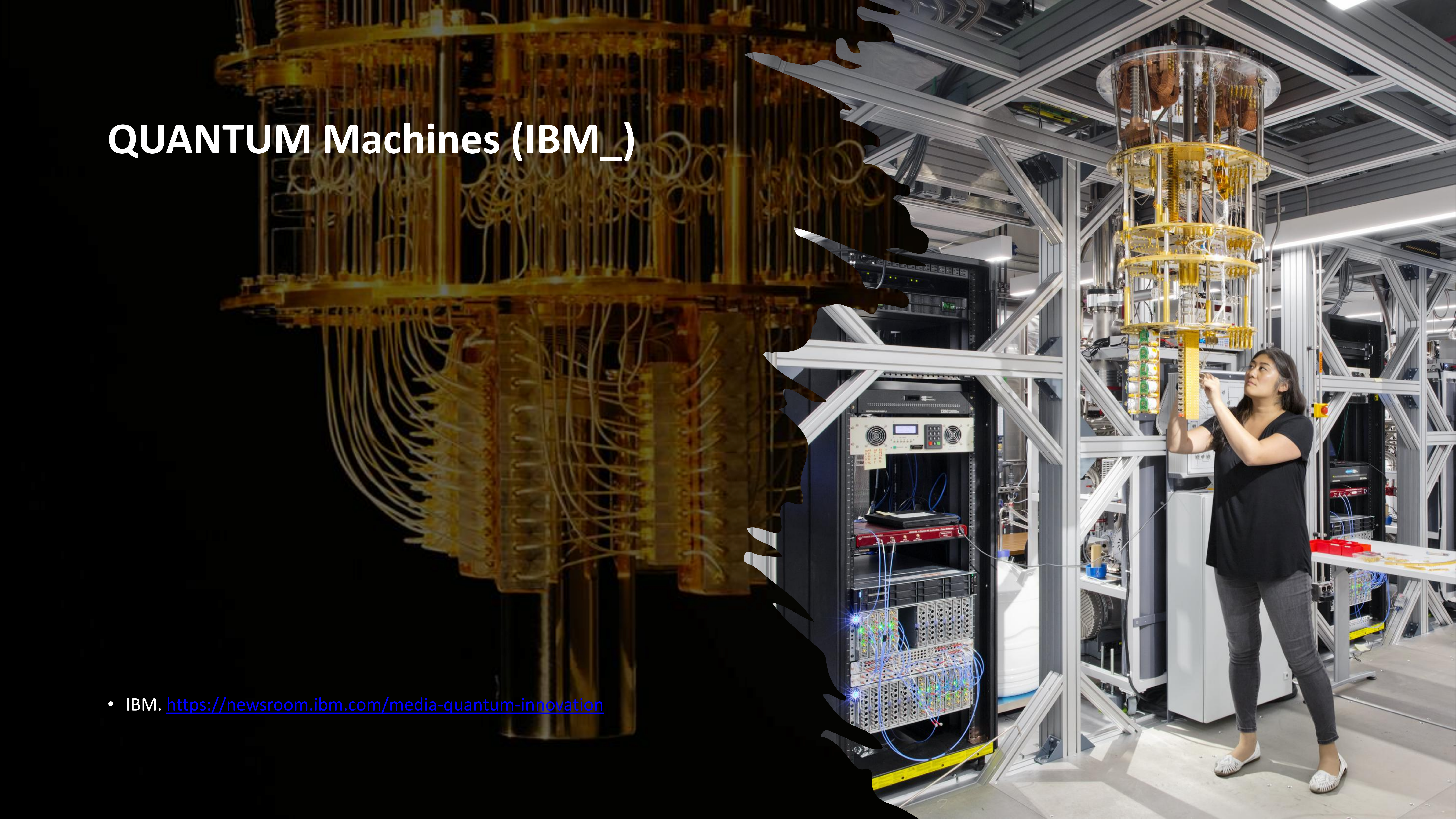
  which are currently used in modern cryptography.



QUANTUM Machines

FT: Crypto industry braces for quantum computing threat. LINK:
https://www.ft.com/content/99c1c1e7-1a1c-479c-9fc8-e21aea5c3f0e?syn-25a6b1a6=1

QUANTUM Machines (IBM_)

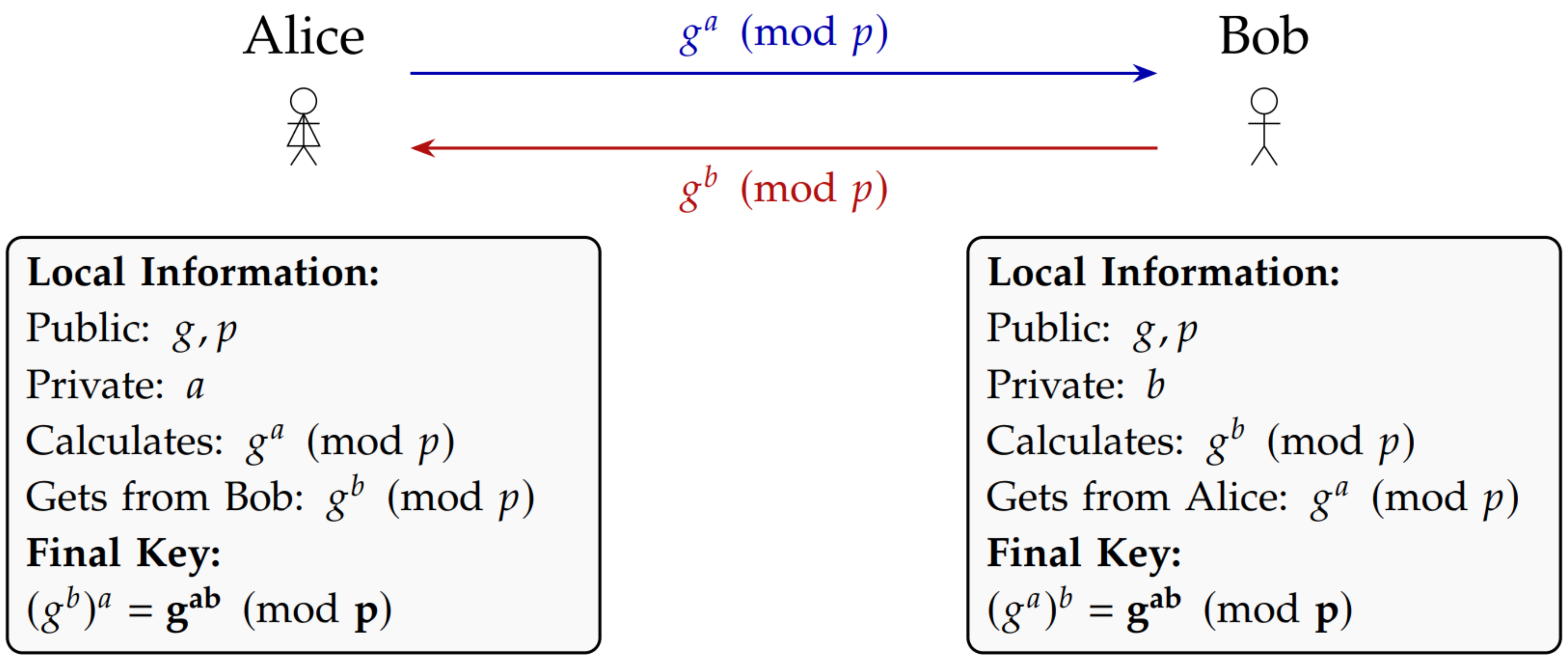


- IBM. https://newsroom.ibm.com/media-quantum-innovation

The simple child asks, “what is quantum computing?” You answer: “it’s a strange new way of harnessing nature to do computation, one that dramatically speeds up certain tasks, but doesn’t really help with others.”

https://scottaaronson.blog/

2 Fundamental Problems that we base A LOT of our modern security on: Discrete Log



https://www.scottaaronson.com/tcs.pdf <- Image source

2 Fundamental Problems that we base A LOT of our modern security on: … and Factoring

The **fundamental theorem of arithmetic** (or **unique factorization theorem** / **prime factorization theorem)**, states that every integer greater than 1 is either prime or can be represented uniquely as a product of prime numbers.

https://en.wikipedia.org/wiki/Fundamental_theorem_of_arithmetic

$$12 = 3 * 3 * 2$$

$$183 = 3 \ * 61$$

929 = ?

2 Fundamental Problems that we base A LOT of our modern security on:
… and Factoring like RSA

An RSA-like Problem

Select large prime p,
Select large prime q,
Compute **N = p*q**

Challenge

Given only N,
Find its prime factors

QUANTUM 2

Quantum computing poses a threat to classical (everyday, modern) cryptography.

WHY?

Cryptographic protocols like RSA, which rely on the difficulty of factoring large semiprimes, and Elliptic Curve Cryptography (ECC), which depend on the complexity of solving discrete logarithms, are vulnerable to quantum attacks.

- Shor's algorithm reduces the time needed to break these cryptographic schemes.
- A quantum computer powerful enough to run Shor’s algorithm could break RSA or Elliptic Curve Crypto in minutes or hours
- These are tasks that would otherwise require millions of years with classical computers.

QUANTUM 3

Blockchain relies heavily on cryptography for signatures and hashing. How vulnerable is it to quantum attacks?

Elliptic Curves:

- Most blockchains, including Bitcoin and Ethereum, depend heavily on Elliptic Curve Cryptography for generating addresses, key-pairs, and digital signatures.

Risks:

- Quantum computers could recover private-keys from public addresses, allowing attackers to forge digital signatures, which could mean what?
- Although blockchain uses hash functions like SHA-256, which are somewhat resistant to quantum attacks, the reliance on ECC poses an immediate risk.
- What is the risk to SHA-256?

Long-term Implications:

- Without updates or new standards, blockchain security could be severely compromised.

QUANTUM 4

Post-Quantum Cryptography
Given the threat, researchers have developed quantum-resistant cryptography, known as Post Quantum Cryptography, or PQC."

What is PQC?

- Post-Quantum Cryptography involves algorithms specifically designed to be secure against *both* **classical** and **quantum computing** threats.

Types of PQC:

- The main categories include lattice-based, hash-based, and code-based cryptography, among others.
- Lattice-based methods like CRYSTALS-Kyber (key exchange) and CRYSTALS-Dilithium (digital signatures) are being standardized by NIST.

Why PQC Works:

- These algorithms rely on mathematically & computationally hard problems considered to be extremely difficult even for quantum computers.

Transition:
"Let’s finish by exploring quantum-secure key agreement, which we’ll soon implement practically."

QUANTUM News (2026)

A couple of recent papers have been published with reduced estimates for the number of qbits needed for quantum attacks on elliptic-curve cryptography.

1. https://arxiv.org/abs/2603.28846 Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities: Resource Estimates and Mitigations, April 2026
2. https://arxiv.org/abs/2603.28627 Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits, March 2026

This could be a concern, as what if we don’t see the gradual speed advancements, but rather huge jumps in capability? Could we attack Bitcoin-like systems with “on-spend attacks”.

Paper 1 points out that these systems are vulnerable in that there is no 2-factor auth, once the key is recovered (stolen), your funds are vulnerable. This is unlike many other systems which have extra security mechanisms.

QUANTUM

Practical...

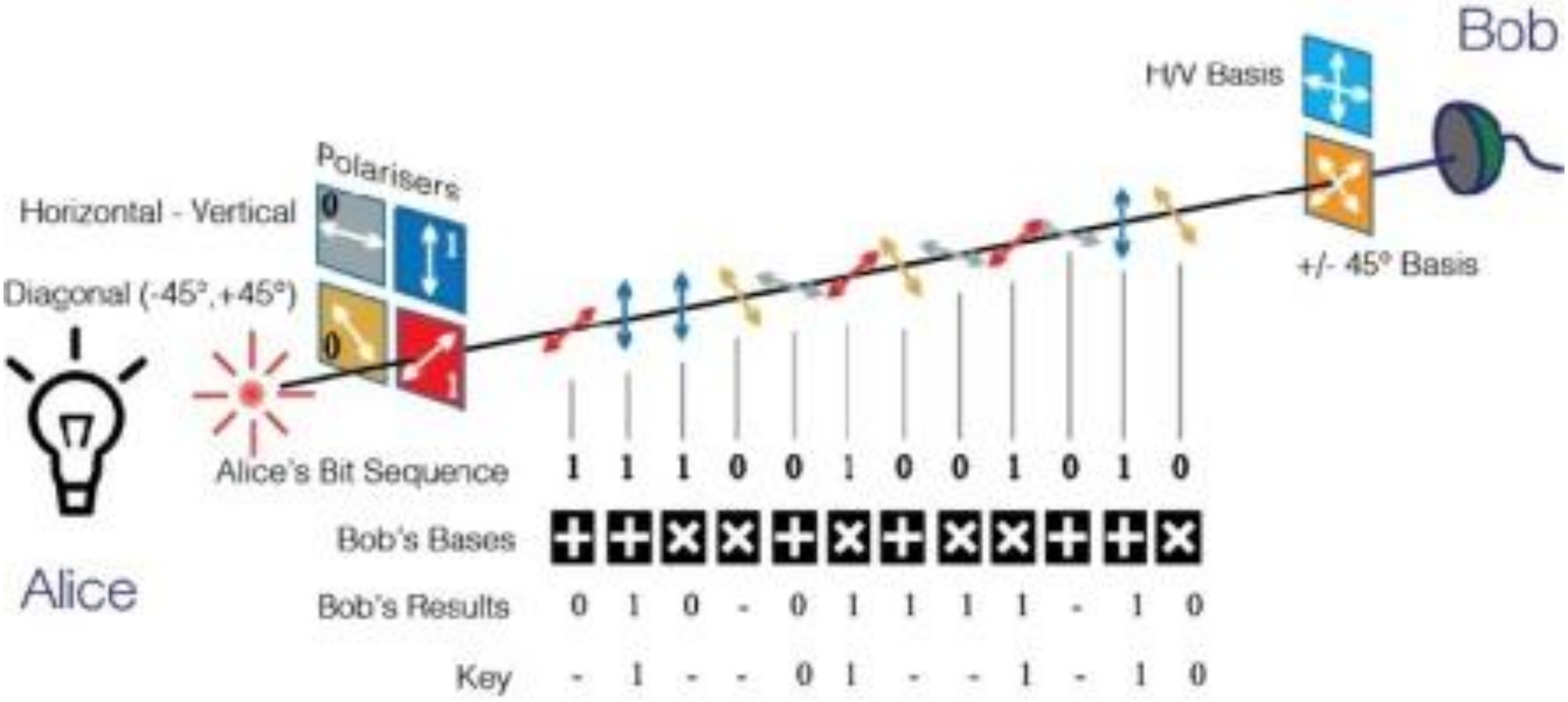
QUANTUM Key Distribution

Quantum Key Distribution (QKD), Unbreakable Security Through Physical Laws

Quantum Key Distribution (QKD) uses individual photons to transmit encryption keys between two parties. Any attempt to intercept disturbs their quantum states due to the measurement principle, that observing a quantum system changes it. This disturbance creates detectable errors in the transmitted data, revealing any eavesdropping attempt.

The security comes from two quantum laws: the Heisenberg Uncertainty Principle prevents extracting complete information without introducing errors, and the No-Cloning Theorem makes it impossible to copy unknown quantum states. Unlike traditional cryptography based on mathematical complexity, QKD's security is guaranteed by physics, and so if no errors are detected, the key is provably secure.

THIS IS VERY DIFFERENT FROM KEY DISTRIBUTION SYSTEMS WE HAVE TODAY



Source: https://qt.eu/quantum-principles/communication/quantum-key-distribution-qkd